2-1 Flowcharts

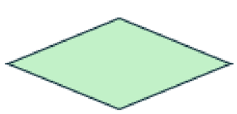
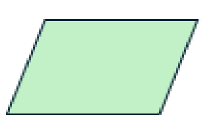
A flowchart is a graphical representation of an algorithm or of a portion of an algorithm. Flowcharts are drawn using symbols.

One reason why flowcharts are so popular is because they use standardized shapes that almost anyone can instantly recognize.

Here are the five most common shapes used in a flowchart.

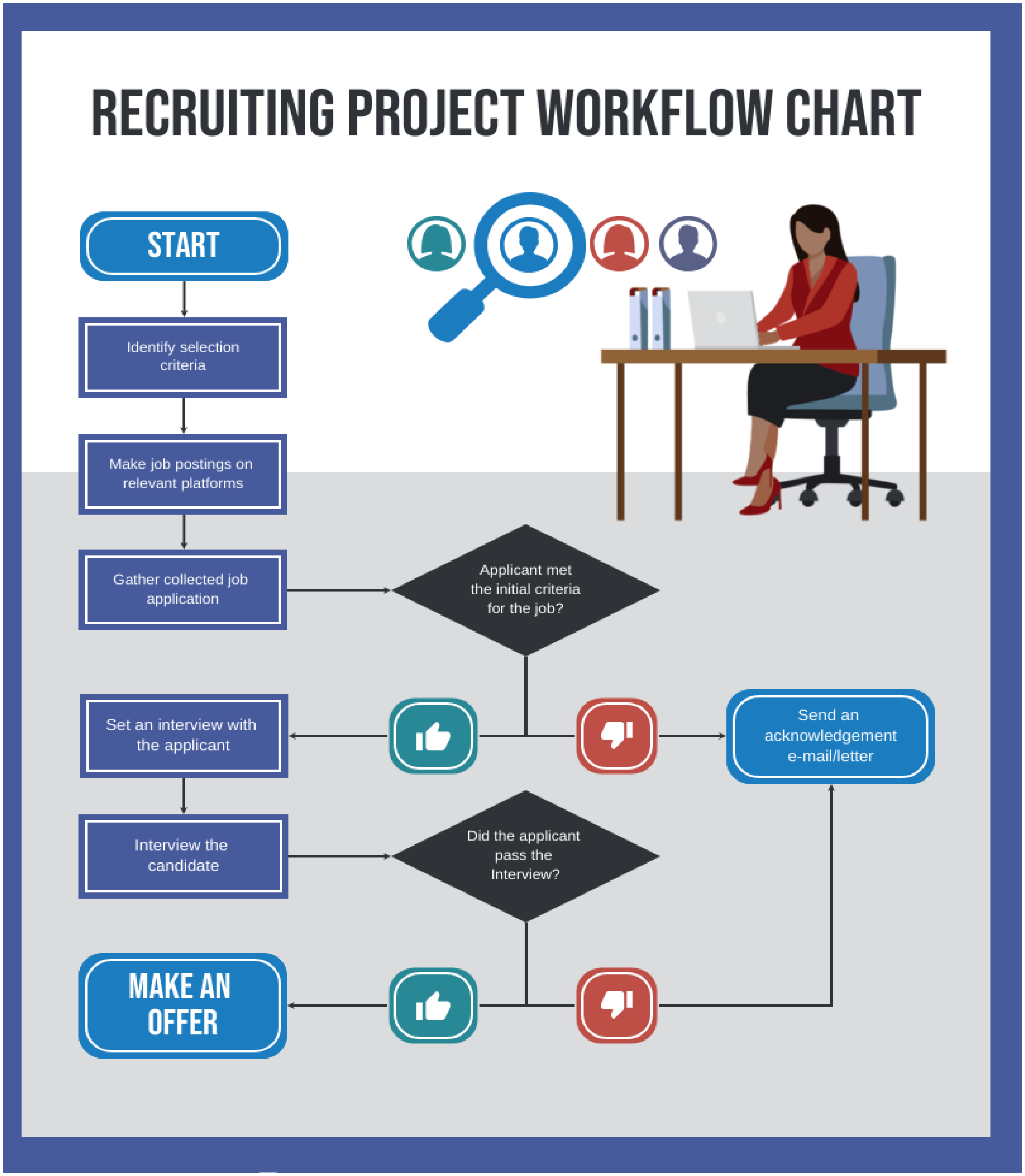
- **Oval** (Terminal symbol)
- **Rectangle** (Process symbol)
- **Arrow** (Arrow Symbol)
- **Diamond** (Decision symbol)
- **Parallelogram** (Input/output symbol)

The main symbols used to draw a flowchart are shown in following figure.

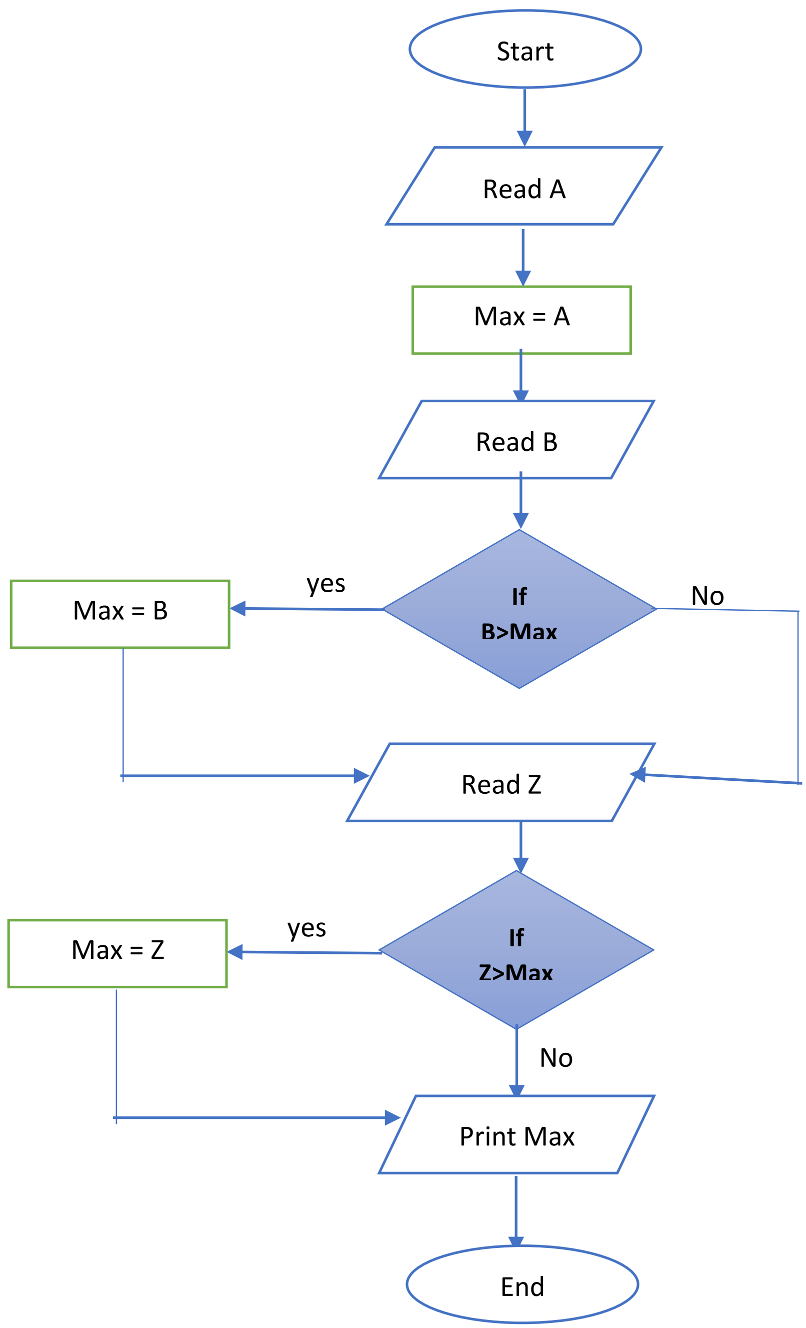
Symbol	Name	Function
	Oval	Represents the start or end of a process
	Rectangle	Denotes a process or operation step
	Arrow	Indicates the flow between steps
	Diamond	Signifies a point requiring a yes/no
	Parallelogram	Used for input or output operations

Example (1):

Draw flowchart for job application using recruiting workflow chart.



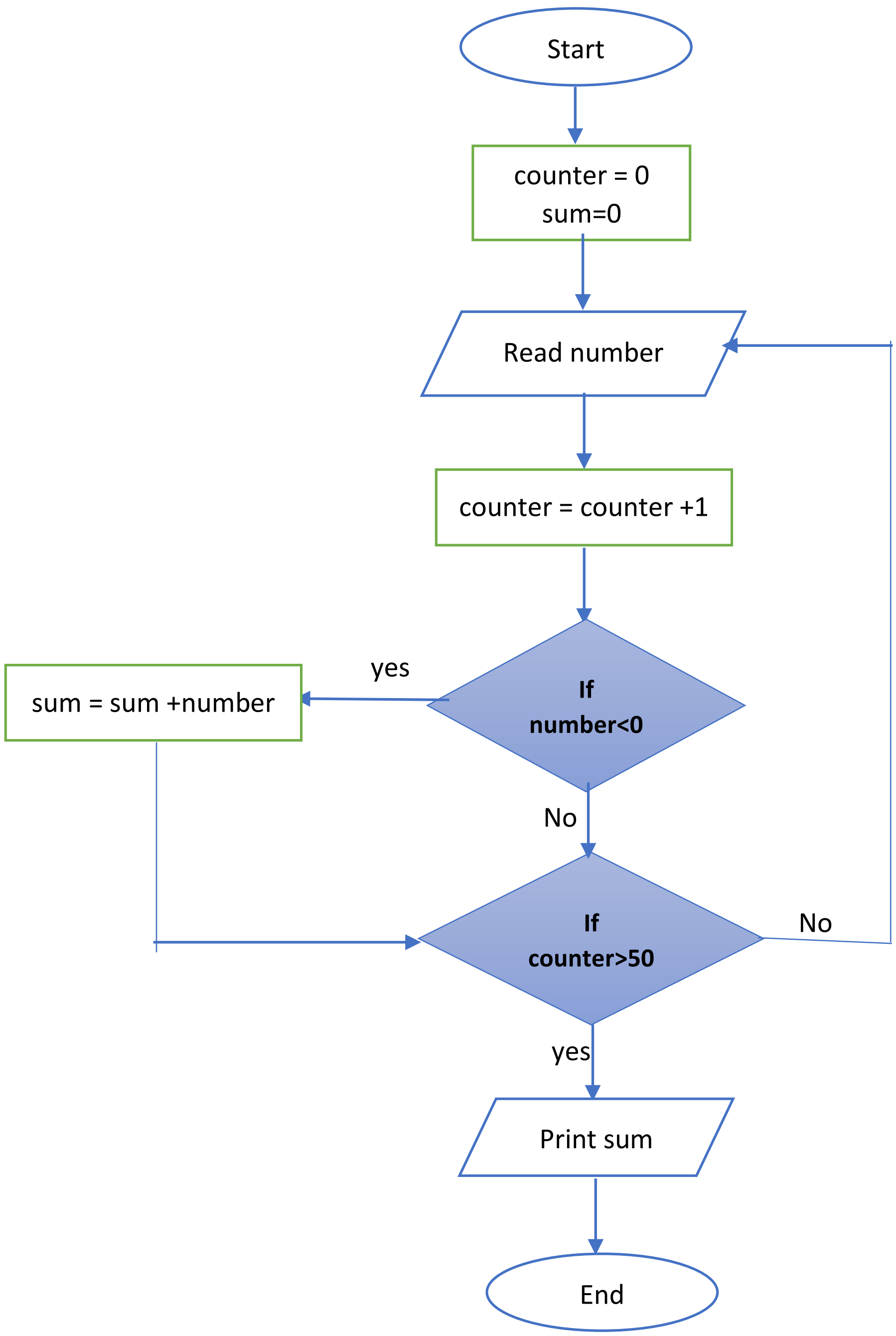
Example (2):
Draw a flowchart to read 3 numbers: A, B and C and print the largest number of them.
Sol.
Flowchart to find the largest of three numbers, A, B, and C starts by inputting the three numbers, then compares them in a series of nested decisions. First, it checks if A is greater than B. If true, it compares A with C to determine if A is the largest; if false, it compares B with C to find the largest.



Example 2:
Draw the flowchart required to find the sum of negative numbers among 50 numbers entered by the user.

Sol.

Create three variables the first is called "*counter*" to held entered number and the second called "*sum*" to compute summation of the negetive numbers the third is called "*number*" to read the inputs.



Homework (1)

Write an algorithm and flowcharts for the following:

a. Sum the even numbers for n numbers.
b. Display numbers from Oto 10.
c. The multiplication of 10 numbers.